Journal of Proteomics 315 (2025) 105412

Contents lists available at ScienceDirect

Journal of Proteomics

journal homepage: www.elsevier.com/locate/jprot

Aberrant oxidative modifications of neutrophil myeloperoxidase in anti-neutrophil cytoplasmic antibody-associated vasculitis

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ARTICLE INFO

Keywords:
ANCA-associated vasculitis
Autoantibody
LC-MS
Myeloperoxidase
Oxidative modifications
SWATH

ABSTRACT

Anti-neutrophil cytoplasmic antibodies directed to myeloperoxidase (MPO-ANCA) are key molecules in the pathogenesis of ANCA-associated vasculitis (AAV), however, the mechanisms of autoantibody production have not been elucidated. We hypothesized that an aberrant PTM occurs in the MPO of MPO-ANCA-positive AAV (MPO-AAV), which induces immune responses to self MPO. To test this, we purified MPO proteins from neutrophils of 8 patients with MPO-AAV and 8 healthy subjects, digested them with trypsin, and comprehensively quantified PTMs of the MPO peptides using the sequential window acquisition of all theoretical fragment ion spectra (SWATH) method of LC-MS. Among the 1034 detected MPO peptides, 38 peptides were increased in the patients with MPO-AAV relative to the healthy subjects, whereas 10 peptides were decreased in the patients ($p < 0.05$). Interestingly, oxidative modifications were found in 11 of the 38 increased peptides (1.14- to 3.29-fold), but not in the decreased peptides. These included oxidation of Met577, Phe686, Met688 and Met719, dioxidation of Met409, Phe605, Trp679 and Met719, and kynurenylation of Trp255. Conversely, glycosylation was detected in 4 of the 10 decreased peptides (−1.32- to −2.32-fold), but not in the increased peptides. They were *O*-type glycans at Ser357 and Ser731, and *N*-type glycans at Asn355 and Asn729. In animal experiments, immunization of mice with in vitro oxidized or unoxidized mouse MPO (mMPO) showed that not only anti-oxidized mMPO antibodies but also anti-unoxidized mMPO antibodies were preferentially produced in the oxidized mMPO-immunized mice relative to the unoxidized mMPO-immunized mice (anti-oxidized mMPO antibodies, 6/8 vs 1/9, $p < 0.05$; anti-unoxidized mMPO antibodies, 4/8 vs 0/9, $p < 0.05$). Our results suggest that the increased oxidative modifications of MPO in MPO-AAV may break immune tolerance and trigger the MPO-ANCA production.

Significance: AAV is a systemic and refractory disease that causes life-threatening multi-organ involvement such as necrotizing glomerulonephritis and lung hemorrhage. MPO-ANCA is an autoantibody that plays a key role in the pathogenesis of AAV. Therefore, elucidation of the mechanism of MPO-ANCA production is crucial to overcoming this disease. In this study, we applied a SWATH-MS analysis to the detection of aberrant PTMs, and found increased oxidative modifications of neutrophil MPO in patients with MPO-AAV for the first time.

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https://doi.org/10.1016/j.jprot.2025.105412
Received 16 November 2024; Received in revised form 20 February 2025; Accepted 21 February 2025
Available online 22 February 2025
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Immunization of in vitro oxidized MPO induced autoantibodies to the intact unoxidized MPO, suggesting that the increased oxidative modifications of MPO may break the immune tolerance in MPO-AAV. This study suggests a novel trigger mechanism for MPO-ANCA production.

1. Introduction

Anti-neutrophil cytoplasmic antibody (ANCA)-associated vasculitis (AAV) is a systemic and refractory disease, characterized by necrotizing small to medium vessel vasculitides [1,2]. Patients with severe AAV suffer from life-threatening multi-organ involvements such as necrotizing glomerulonephritis and lung hemorrhage [2–4]. AAV consists of 3 distinct disorders: microscopic polyangiitis (MPA), granulomatosis with polyangiitis (GPA) (formerly known as Wegener's granulomatosis), and eosinophilic granulomatosis with polyangiitis (EGPA) (formerly known as Churg-Strauss syndrome).

ANCA, a characteristic autoantibody for AAV consists of 2 major subtypes: namely one against myeloperoxidase (MPO-ANCA) and the other against leukocyte proteinase 3 (PR3-ANCA). MPO-ANCA is mainly detected in patients with MPA (55–90 %) or EGPA (20–40 %) and less frequently detected in patients with GPA (20–30 %). On the other hand, PR3-ANCA is mainly detected in patients with GPA (65–75 %) and less frequently detected in patients with MPA (20–30 %) or EGPA (<5 %) [3,5,6].

MPO is an enzyme that catalyzes the production of hypochlorous acid (HClO) from hydrogen peroxide (H_2O_2) and chloride ion (Cl^-). MPO is normally located in the azure granules of neutrophils, however, it is also located in the plasma membrane of neutrophils primed by TNFα and other factors [7]. Binding of MPO-ANCA to MPO on the plasma membrane of neutrophils on the vessel wall induces degranulation, the production of ROS, and the release of inflammatory cytokines (e.g. IL-8 and IL-1β) in neutrophils [8–10], resulting in vascular endothelial cell injury [11,12]. Notably, recombination-activating gene 2-knockout mice injected with anti-MPO antibodies or splenocytes producing anti-MPO antibodies develop systemic necrotizing vasculitis and necrotizing glomerulonephritis [13]. These reports indicate that MPO-ANCA is involved in the pathogenesis of AAV. However, the mechanism of autoantibody production against MPO remains unclear.

Alteration of PTM of self-proteins triggers the production of autoantibodies. In rheumatoid arthritis, citrullination of proteins such as fibrinogen and filaggrin serves as a novel epitope, which leads to the production of anti-citrullinated peptide antibodies [14–16]. In Goodpasture syndrome, it is suggested that the hexameric structure of type IV collagen in the basement membrane is altered by ROS, which exposes cryptic epitopes hidden within the hexamers to the immune system, resulting in the production of anti-glomerular basement membrane (GBM) antibodies [17]. In autoimmune diseases, it is well known that intra- and inter-molecular epitope spreading of T cells and B cells occurs according to the disease progression, leading to autoantibody production [18–22]. Based on the above reports, we hypothesized that an aberrant PTM occurs in neutrophil MPO in patients with MPO-ANCA-positive AAV (MPO-AAV), which is involved in the production of MPO-ANCA.

To test this hypothesis, we comprehensively quantified the PTMs of MPO proteins purified from neutrophils of patients with MPO-AAV and healthy subjects, using sequential window acquisition of all theoretical fragment ion spectra-mass spectrometry (SWATH-MS) [23] (Fig. 1). Oxidation, glycosylation, and phosphorylation of MPO have been reported previously (Supplementary Table 1) [24,25]. In this study, we found that some oxidatively modified MPO-derived peptides increased in the results of MPO-AAV patients relative to healthy subjects. Furthermore, mice immunized with murine MPO (mMPO) with oxidative treatment developed antibodies against unoxidized mMPO as well as antibodies against oxidized mMPO. These results suggest that the production of MPO-ANCA may be triggered by increased oxidative modifications of MPO in MPO-AAV. This study suggested a novel mechanism of MPO-ANCA production which may play an important role in the pathogenesis of AAV.

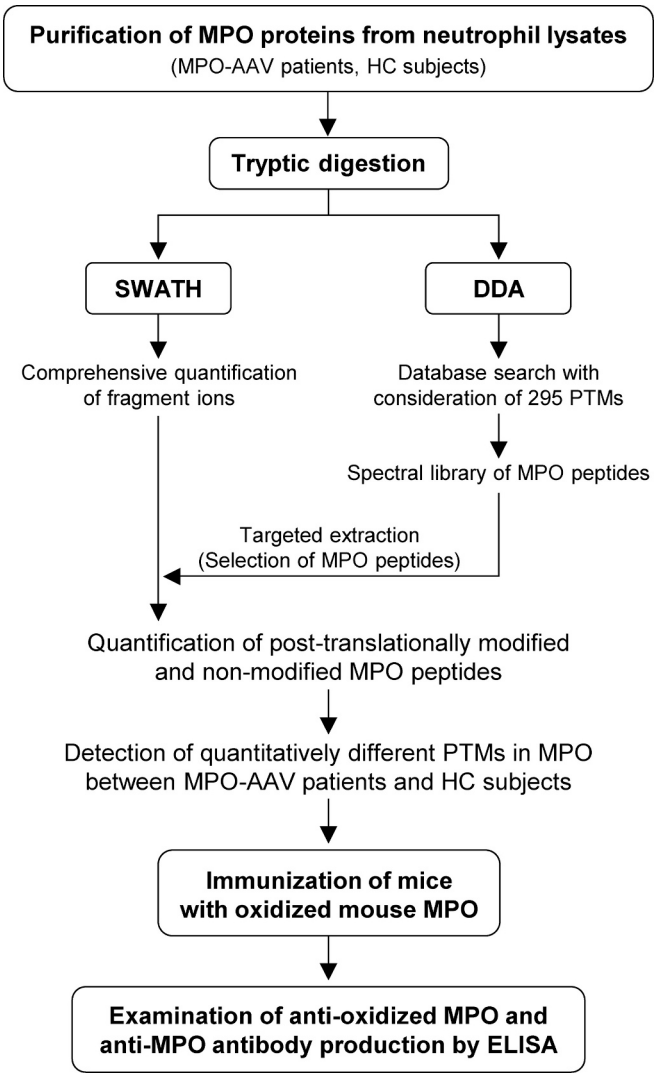


Fig. 1. Study design.
MPO proteins were purified from neutrophil lysates derived from 8 MPO-AAV patients and 8 HC subjects. Tryptic digests of the purified MPO proteins were analyzed by LC-MS/MS in DDA and SWATH modes. The DDA data were subjected to a database search of 295 PTMs. The results obtained in DDA mode were used as a spectral library of MPO peptides. For target extraction, fragment ions of MPO peptides were selected from the SWATH data based on the library information, and their ion intensities in extracted ion chromatography (XIC) were calculated. The obtained ion intensities were compared between the MPO-AAV patients and the HC subjects to detect quantitatively different MPO-PTMs. Since the representatively increased PTM in the MPO-AAV patients was the oxidative modification, mouse MPO oxidized in vitro was used to immunize mice to evaluate the role of PTM in the production of anti-MPO antibodies. To achieve this, anti-oxidized MPO antibodies and anti-MPO antibodies in the plasma from mice immunized with either oxidized or unoxidized MPO were measured by ELISA.

2. Materials and methods

2.1. Clinical samples

Peripheral blood samples were collected from 8 patients with MPO-AAV (mean age ± SD, 76.0 ± 9.7 years; male: female, 4: 4) and 8 healthy control (HC) subjects (mean age ± SD, 74.0 ± 6.4 years; male: female, 4: 4) (Table 1). Diagnoses of vasculitides were made according to the criteria [26]. This research was approved by the Ethics Committees of St. Marianna University School of Medicine (approval number 3861) and Kindai University (approval number H30–1-004), and was registered with the University Hospital Medical Information Network Clinical Trials Registry (UMIN 000036270). The study protocol conformed to the ethical guidelines of the Declaration of Helsinki revised in 2013. Written informed consent was obtained from all participants. All the procedures were performed in compliance with relevant laws and the institutional guidelines.

2.2. Isolation of polymorphonuclear cells and purification of MPO

Polymorphonuclear cells (PMNs), isolated from peripheral blood samples by density-gradient centrifugation using Ficoll-Paque (Cytiva, Tokyo, Japan), were lysed and stored at −80 °C until use [27]. Glycoproteins including MPO were purified from PMN lysates using a Con A column (HiTrap™ Con A 4B; Cytiva, Tokyo, Japan) [28]. MPO-containing fractions were selected by western blotting with rabbit anti-human MPO polyclonal antibodies (Meridian Life Science, Memphis, TN, USA). The MPO proteins in each of the selected fractions were quantified by comparing its signal intensity to that of a 10 ng standard MPO protein (Merck, Darmstadt, Germany). The total amount of purified MPO proteins in each patient or subject was calculated by summing the quantified values of all selected fractions.

2.3. Tryptic digestion of proteins

Tryptic digestion of the protein samples containing MPO was performed as previously described [29]. Briefly, the Con A column eluate containing 2.55 μg of MPO, mixed with an equal volume of ice-cold 30 % TCA, was incubated on ice for 30 min to precipitate proteins. The precipitates were washed with ice-cold acetone, air-dried, and subjected to reductive alkylation of the thiol groups of the proteins. Proteins in the samples were digested with trypsin at 37 °C for 18 h. To stop the digestion reaction, phosphoric acid and formic acid were added to the samples to final concentrations of 0.73 % and 0.15 %, respectively. Peptides produced by digestion, desalted using a ZipTip μC18 (Merck), were completely dried in a centrifugal concentrator. The resulting peptide powder was dissolved in 0.1 % formic acid (Solution A) and stored at −30 °C until use.

2.4. LC-MS analysis

An LC-MS analysis was performed on a Triple TOF5600+ system equipped with a NanoSpray III ion source and an Ecksigent nanoLC system (SCIEX, Framingham, MA, USA). Fractionation of peptide samples was performed as follows. Solution A containing 0.5 μg of the peptides was loaded onto a C_{18} column (ChromXP C18-CL 3 μM 120 A, 75 μm × 150 mm; SCIEX). Then the peptides were separated at a flow rate of 300 nL/min using a binary buffer system using solution A and acetonitrile with 0.1 % formic acid (solution B). The elution gradient at 2 % solution B was increased gradually to 90 % over 125 min (2–5 % in 5 min, 5–20 % in 95 min, 20–32 % in 20 min, and 90 % in 5 min). The eluted peptides were analyzed on a mass spectrometer operated in a data-dependent analysis (DDA) mode or a SWATH-MS mode as described below.

2.5. Generation of MPO peptide spectral library in a DDA mode

To generate a spectral library of MPO-derived peptides, MS/MS spectra of MPO peptides were comprehensively acquired in DDA mode. In the DDA mode, the MS analysis was first performed in the *m/z* range of 400–1250 with an integration time of 0.25 s. A maximum of 20 MS/MS spectra was acquired in the *m/z* range of 100–1600 with an integration time of 50 msec for each MS spectrum. Each sample was analyzed in 2 technical replicates. The equipment parameters were set according to Yamagata's method [29]. All DDA data were subjected to a database search against UniProt *Homo sapiens* protein data (http://www.uniprot.org/) using Protein Pilot™ Software 4.5 (SCIEX). Trypsin was selected as the digestion enzyme and methanethiosulfonation was selected as the cysteine alkylation method in the database search. All 295 modifications available in the Protein Pilot™ software program were considered. Peptides identified with ≥95 % confidence were considered significant. A spectral library of MPO peptides was generated based on peptide data. Based on the detection of the y- and b-ions in the MS/MS spectra and information of PTMs registered in Unimod (https://www.unimod.org), the reliability of PTM sites specified by the Protein Pilot analysis algorithm was classified into 3 categories as follows: +++, the modified amino acids were identified from the MS/MS spectra; ++, the modified amino acids were not identified from the MS/MS spectra but were identified from the already-known information of amino acids that undergo the PTM types; +, the modified amino acids were not identified from the MS/MS spectra or already known information.

Table 1
Clinical information of the enrolled MPO-ANCA positive AAV patients.

Patients	Age (years)	Sex	Disease durations	MPO-ANCA (U/mL)	PR3-ANCA (U/mL)	CRP (mg/dL)	WBC (10^3/μL)	IgG (mg/dL)	Diagnoses	Treatments
AAV1	82	F	2 m	**>300**	<1.0	**3.39**	5.9	1346	MPA	None
AAV2	82	M	4 y	**47.2**	1.0	**11.14**	**10.5**	903	MPA	PSL 4 mg, CyA 50 mg
AAV3	71	F	15 y	**18.3**	<1.0	**4.57**	6.6	**1892**	MPA	PSL 20 mg,
AAV4	67	F	3 y	**13.9**	<1.0	**8.88**	**17.0**	986	RLV	PSL 10 mg, Bredinin 50 mg
AAV5	87	M	1 m	**>300**	<1.0	**6.10**	**10.5**	1091	MPA	None
AAV6	82	M	2 m	**151.9**	<1.0	**11.39**	**11.1**	901	AAV	None
AAV7	56	F	1 m	**54.8**	<1.0	**4.76**	**12.8**	1605	AAV	None
AAV8	81	M	7 m	**46.9**	<1.0	**4.20**	8.3	**2170**	OMAAV	None

Laboratory data and treatments on the dates peripheral blood samples were obtained are shown.
Reference values: MPO-ANCA, 0–3.5 U/mL; PR3-ANCA, 0–3.5 U/mL; CRP, <0.3 mg/dL; WBC, 4–9 × 10^3/μL; IgG, 760–1800 mg/dL. Values in bold are above the reference ranges.
AAV, ANCA-associated vasculitis; ANCA, anti-neutrophil cytoplasmic antibodies; CRP, C-reactive protein; CyA, cyclosporine A; MPA, microscopic polyangiitis; MPO, myeloperoxidase; OMAAV, otitis media with AAV; PR3, proteinase-3; PSL, prednisolone; RLV, renal limited vasculitis type of AAV; WBC, white blood cells.

2.6. Quantification of MPO peptides in the SWATH-MS mode

In SWATH mode, 50 variable-sized Q1 windows were stepped across the 400–1250 *m/z* range based on the distribution of precursor *m/z* values using SWATH Variable Window calculator Tool ver 1.0 (SCIEX). The collision energy for each window was determined based on the appropriate collision energy for an ion with a 2+ charge centered on the window with a spread of 15 eV. The combination of a 100 msec survey scan in the range of 400–1250 *m/z* and 50 SWATH scans (50 msec per scan) in the range of 100–1500 *m/z* was continuously repeated for 150 min. The obtained SWATH-MS data were analyzed using Peak View 2.1 (SCIEX). We manually corrected analytical errors after automated analyses of 1) the targeted extraction using the retention time of precursor ion and the *m/z* values of precursor and fragment ions registered in the MPO peptide spectral library, and quantification of fragment ions derived from a given MPO peptide; 2) the false discovery rate of the quantification results; and 3) a gap value that indicates the percentage of missing values. The peak areas of the extracted ion chromatograms (XIC) of the 5–14 selected fragment ions derived from the same precursor ion were summed up for the quantification of the precursor ion. The quantitative values of individual MPO peptides were normalized by dividing them by the total quantitative value of all detected MPO peptides in each patient or subject, according to Lambert's method [30]. Thereafter, the normalized quantitative values were used for the comparison between the MPO-AAV group and the HC group.

2.7. H_2O_2-treatment of mMPO

mMPO was oxidized using H_2O_2 and $FeCl_2$. Briefly, recombinant mMPO (R&D systems, Minneapolis, MN) was dissolved in 0.2 M sodium acetate buffer (pH 4.7) containing 4 % CHAPS and 1 M NaCl to prepare 2.5 μg/μL mMPO solution. The final concentrations of 0.2 mM H_2O_2 and 0.01 mM $FeCl_2$ were added to the mMPO solution (H_2O_2-treated mMPO). An mMPO solution without H_2O_2 and $FeCl_2$ was used as a control (non-treated mMPO). These reaction mixtures were incubated at 37 °C for 30 min and subjected to tryptic digestion and desalting, similar to the treatment for human MPO, as described above. The produced peptides, dissolved in 0.1 % formic acid, were stored at −30 °C until use. LC-MS and SWATH-MS were performed using 3.0 μg of H_2O_2-treated or non-treated mMPO, similar to the analysis for human MPO. The obtained data were subjected to peptide identification using the UniProt *Mus musculus* protein database.

2.8. Immunization of mice with the oxidized mMPO and titration of anti-MPO antibodies

Ten micrograms of H_2O_2-treated mMPO were mixed with an equal weight of complete Freund's adjuvant (CFA) or incomplete Freund's adjuvant (IFA). H_2O_2-treated mMPO with CFA and H_2O_2-treated mMPO with IFA were administered intraperitoneally to 8-week-old female C57BL/6 J Kwl mice (B6 mice) (Kiwa Laboratory Animals Co., Ltd., Wakayama, Japan) on day 0 and 21, respectively. As a negative control, non-treated mMPO was similarly mixed with CFA or IFA and administered to B6 mice. Furthermore, 10 μg of BSA (FUJIFILM Wako Pure Chemical Corporation, Osaka, Japan) mixed with CFA or IFA was similarly administered to B6 Mice. Blood samples were collected from the retro-orbital plexus on days 0, 15, and 30, and from the hearts on day 34. Anti-MPO antibodies were titrated using an ELISA, as follows. Each well in a 96-well microplate was coated with 50 ng H_2O_2-treated mMPO, or 50 ng non-treated mMPO. Mouse plasma was diluted 5000-fold and used as primary antibody. Antibodies against the coated proteins were titrated using KPL TMB Microwell Peroxidase Substrate System (SeraCare Life Sciences, Inc., Milford, MA, USA). Titers beyond the mean titer +3SD of BSA-immunized mice on day 34 were defined as antibody positive. The experiments were carried out in 2 groups (1st group, H_2O_2-treated mMPO, $n = 4$; non-treated mMPO, $n = 5$; 2nd group, H_2O_2-treated mMPO, $n = 4$; non-treated mMPO, $n = 4$). Animal experiments were approved by the Animal Experimentation Subcommittee of the Faculty of Biology-Oriented Science and Technology, Kindai University (approval number, KABT-31-012). All animal experiments complied with the ARRIVE guidelines and were performed in accordance with the National Research Council's Guide for the Care and Use of Laboratory Animals.

2.9. Database analysis of PTMs of MPO

Glycosylation in human MPO was analyzed using GlyGen (data ver. 2.4.1; [31]). PTMs other than glycosylation were analyzed using PhosphoSitePlus® (v6.7.4; [32]). Three-dimensional structural maps of human MPO were generated using Waals ver 2.0 software (Altif Laboratories Inc., Tokyo, Japan) with Protein Data Bank (ID of MPO, 1CXP) for a three-dimensional structural model of MPO.

2.10. Statistical analysis

Principal component analysis (PCA) was performed using SIMCA ver. 14 (Sartorius AG, Göttingen, Germany) to detect outliers in the SWATH-MS results. The quantified SWATH-MS data were compared between the MPO-AAV group and the healthy group and between H_2O_2-treated mMPO and non-treated mMPO using Student's *t*-test. The significance of differences in anti-MPO antibody production between the H_2O_2-treated mMPO-immunized mice and the non-treated mMPO-immunized mice was evaluated using Fisher's exact test.

3. Results

3.1. Comprehensive exploration of aberrant MPO-PTMs in the MPO-AAV patients

3.1.1. Detection of PTMs in MPO peptide spectral library of the MPO-AAV patients and HC subjects

To detect aberrant PTMs of MPO in MPO-AAV, we first comprehensively analyzed the PTMs of neutrophil MPO in MPO-AAV patients and HC subjects. MPO proteins were purified from neutrophil lysates derived from 8 MPO-AAV patients and 8 HC subjects (Supplementary Fig. 1). Tryptic digests of MPO proteins were analyzed by LC-MS/MS in the DDA mode. The MPO-AAV patients and the HC subjects had similar numbers of identified MPO peptides (MPO-AAV patients, 2802 ± 374; HC subjects, 2459 ± 266) and similar sequence coverages for MPO (MPO-AAV patients, 75.5 ± 1.8 %; HC subjects, 75.6 ± 1.0 %) (Supplementary Table 2 and Supplementary Fig. 2). From the DDA results, 1034 MPO peptides containing 43 types of PTMs were identified (Supplementary Table 3).

The 43 types of PTMs were found at the 436 amino acid sites of MPO (Supplementary Table 3). Among these, oxidative modifications were most frequently detected at 105 amino acid sites. In particular, the oxidation of Met (17 sites) and Trp (8 sites), dioxidation of Trp (9 sites) and Met (8 sites), and conversion of Trp to Kyn (9 sites) were frequently observed. Deamidation was the second most frequent PTM at the 75 amino acid sites. Deamidation of Asn (30 sites), Gln (23 sites), and Arg (22 sites) was observed. Other enzymatically-catalyzed PTMs of glycosylation, phosphorylation, and methylation were also detected. All 5 *N*-linked glycosylation sites (Asn323, Asn355, Asn391, Asn483, and Asn729) and 2 *O*-linked glycosylation sites (Ser322 and Ser481) of MPO registered in the glycan database (GlyGen) were found in our results. Furthermore, novel *N*-linked (at Asn358) and *O*-linked (at Ser340, Ser357, Ser485, Thr503, Ser731, and Thr732) glycosylation sites were identified in this study (Supplementary Fig. 3). Similarly, novel phosphorylation sites (Thr613 and Thr707) that were not registered in the PTM database (PhosphoSitePlus®) were also identified (Supplementary Fig. 3). N-terminal carbamoylation and β-methylthiolation of cysteine were also observed.

In the DDA mode, the types and sites of MPO-PTMs identified in the MPO-AAV patients were almost identical to those identified in the HC subjects, suggesting that there were no specific PTMs in MPO-AAV (Supplementary Table 3). Thus, the results obtained in this DDA analysis were used as a spectral library of MPO peptides, and subjected to a comprehensive quantitative comparison of MPO peptides between the MPO-AAV patients and the HC subjects using SWATH.

3.1.2. Quantitative comparison of post-translationally-modified MPO peptides between the MPO-AAV patients and the HC subjects

Next, to find quantitatively-aberrant PTMs of MPO in MPO-AAV, the post-translationally-modified MPO peptides were quantified in SWATH mode. In the 1034 post-translationally-modified MPO peptides identified by the DDA mode, precursor ions of 458 peptides were quantified in all samples using the SWATH mode. Because SWATH data analysis algorithms in the Peak View software program exclude peptides that are not detected in even 1 of the 16 samples, peptides detected in MPO-AAV patients but not in HC subjects (or vice versa) could not be analyzed. To avoid this, we manually confirmed that no MPO peptide was detected in either the MPO-AAV patients or the HC subjects. Thus, the SWATH data for the 458 peptides were selected for subsequent analysis. The quantitative values of individual MPO peptides were normalized using the total quantitative value of the 458 MPO peptides in each patient or subject.

To examine whether the PTMs of MPO could discriminate the MPO-AAV patients from the HC subjects, the normalized quantitative values of the 458 peptides were subjected to PCA (Supplementary Fig. 4). As a result, PCA did not clearly discriminate between the MPO-AAV patients and the HC subjects. In contrast, 2 samples, 1 derived from AAV7 and the other from HC6, were shown as outliers in the PCA result. Therefore, we excluded the results of 2 samples from the subsequent comparison analysis.

The normalized quantitative values of the 458 trypsin-digested MPO peptides were compared between the MPO-AAV group ($n = 7$) and the HC group ($n = 7$). As a result, 48 peptides showed significantly different quantitative values between the 2 groups ($p < 0.05$) (Table 2 and Supplementary Table 4). Of these, 38 peptides were increased and the remaining 10 were decreased in the MPO-AAV group (Table 2). Interestingly, oxidative modifications were found in 11 of the increased 38 peptides, but not in the decreased 10 peptides. The 11 increased peptides included oxidative modifications at 8 amino acid sites (Trp255, Met409, Met577, Phe605, Trp679, Phe686, Met688, and Met719; Table 2, and Supplementary Fig. 5). The oxidative modifications were oxidation of Met577, Phe686, Met688 and Met719; dioxidation of Met409, Phe605, Trp679 and Met719; and kynurenylation of Trp255 (Fig. 2, Supplementary Fig. 5, and Supplementary Fig. 6). The top 3 peptides in Table 2, which increased approximately 3-fold in the MPO-AAV group, included the oxidation of Met719, Met688, and Phe686. In contrast, glycosylation was found in 4 of the 10 peptides decreased in the MPO-AAV group, but not in the 38 increased peptides. The types of glycosylation were *O*-linked (at Ser357 and Ser731) and *N*-linked (at Asn355 and Asn729). Two of the bottom 3 peptides in Table 2, which decreased approximately 2-fold in the MPO-AAV group, included *N*-linked and *O*-linked glycosylations (at Asn355 and Ser357). Deamidation was found in 4 out of 10 decreased peptides (at Asn492, Asn496, and Asn737) and 2 out of 38 increased peptides (at Gln 257 and Asn496) (Table 2). Carbamoylation was found in 26 of the 38 increased peptides in the MPO-AAV group, but not in the 10 decreased peptides.

Following the results of the increased oxidative modifications at eight amino acid sites of MPO in the MPO-AAV group, we examined the spatial positions of these amino acids within the MPO molecule. 3D-analysis of MPO revealed that 5 oxidized amino acids were located on the surface (Met577, Trp679, Phe686, Met688, and Met719), 2 located at the catalytic site containing heme (Trp255 and Met409), and 1 located at the interface of the MPO dimer (Phe605) (Supplementary Fig. 7). In addition, all the glycosylated amino acids (Asn355, Ser357, Asn729, and Ser731) were located on the surface of the MPO molecule [33].

3.2. Induction of anti-MPO antibodies by immunization with oxidatively-modified MPO

3.2.1. Introduction of oxidative modifications into mouse MPO by H_2O_2-treatment

To investigate whether an increase in oxidative modification of MPO is involved in the production of MPO-ANCA, autoantibodies to MPO were measured in mice immunized with mMPO oxidatively-modified using H_2O_2 (H_2O_2-treated mMPO). As the first step, PTMs of H_2O_2-treated mMPO and non-treated mMPO were analyzed using the same LC-MS method as that used for human neutrophil-derived MPO in the DDA and SWATH modes. DDA identified a total of 781 mouse MPO peptides, of which 89 PTMs were detected (Supplementary Table 5). Of the 781 identified peptides, 469 were available for a quantitative analysis using the SWATH mode. The normalized quantitative values of 411 of the 469 peptides differed between the H_2O_2-treated mMPO protein samples and the non-treated mMPO protein samples ($p < 0.05$) (Supplementary Table 6). Of the 33 peptides whose peak areas were $\geq$ 30-fold greater in H_2O_2-treated mMPO relative to non-treated mMPO, 29 peptides included oxidative modifications at Met, Trp, Tyr and Pro (Table 3). As mentioned above, oxidative modifications at 8 amino acid sites were increased in the MPO-AAV group relative to the HC group (see 3.1.2; Table 2 and Supplementary Fig. 7). In the H_2O_2-treated mMPO, the oxidative modifications at 5 of the 8 amino acid sites that corresponded to the 8 highly oxidatively modified sites of MPO in the MPO-AAV patients were increased relative to the non-treated mMPO (Supplementary Fig. 5). Interestingly, the types of oxidative modification at 4 of the 5 amino acid sites were identical between human and mouse MPO proteins, that is, kynurenylation of human Trp255 and mouse Trp229, dioxidation of human Met409 and mouse Met383, dioxidation of human Met577 and mouse Met551, and oxidation of human Met719 and mouse Met693 (Supplementary Fig. 5 and Table 3). The remaining 1 site was human Trp679 with dioxidation and mouse Trp653 with oxidation and kynurenylation. Therefore, a considerable part of the oxidative modifications that increased in the MPO-AAV patients was successfully introduced in H_2O_2-treated mMPO.

3.2.2. Induction of antibodies against H_2O_2-treated and non-treated mMPO by immunization with H_2O_2-treated mMPO

Next, to examine whether autoantibodies to MPO are produced in mice that possess oxidatively modified MPO, antibodies to MPO were measured in mice immunized with H_2O_2-treated mMPO. Specifically, B6 mice were immunized twice with H_2O_2-treated mMPO or non-treated mMPO, and blood samples were collected according to the time course shown in Fig. 3A. Antibody titers against H_2O_2-treated mMPO and non-treated mMPO were measured by ELISA (Fig. 3B-E). On days 15, 30 and 34, one, five, and five of the eight mice immunized with H_2O_2-treated mMPO were positive for antibodies against H_2O_2-treated mMPO, respectively (Fig. 3BD, left). On days 30 and 34, four out of the eight mice immunized with H_2O_2-treated mMPO were also positive for antibodies against non-treated mMPO (Fig. 3CE, left). In contrast, 8 of the 9 mice immunized with non-treated mMPO were negative for antibodies against H_2O_2-treated mMPO and for antibodies to non-treated mMPO during the time course (Fig. 3BC, right; Fig. 3DE, center). The remaining 1 mouse was weakly positive for antibodies against H_2O_2-treated mMPO on days 30 and 34 (Fig. 3B, right) and for antibodies against non-treated mMPO on day 30 (Fig. 3C, right). On day 34, the prevalences of the antibodies against H_2O_2-treated mMPO and of the antibodies against non-treated mMPO in H_2O_2-treated mMPO-immunized mice were significantly higher than those in non-treated mMPO-immunized mice ($p < 0.05$) (Table 4).

Table 2
Quantitatively different MPO peptides between the MPO-AAV group and the HC group.

Confirmed sequence of MPO peptides	Theoretical *m/z* of precursor ion	Precursor charges	Amino acid sites	Focused PTMs				Reliability of the modification sites*	Average ± SD of normalized intensity ($\times 10^{-5}$)		MPO-AAV/HC	*p*-values
				Oxidation	Carbamoylation	Deamidation	Glycosylation		MPO-AAV ($n = 7$)	HC ($n = 7$)		
[CRM]-^{715}NNIFM[Oxi]SNSYPR^{725}	701.31445	2	Met719	✓				+++	13.96 ± 3.00	4.24 ± 2.50	3.29	2.6E-05
			Asn715 (N-term)		✓			+++				
[CRM]-^{678}FWWENEGVFSM[Oxi]QQR^{691}	951.91504	2	Met688	✓				+++	8.20 ± 1.58	2.58 ± 1.45	3.18	1.5E-05
			Phe678 (N-term)		✓			+++				
[CRM]-^{678}FWWENEGVF[Oxi]SMQQR^{691}	951.91504	2	Phe686	✓				+	6.88 ± 2.97	2.52 ± 1.50	2.73	4.6E-03
			Phe678 (N-term)		✓			++				
[CRM]-^{629}KLMEQYGTPNNIDIWMGGVSEPLKR^{653}	973.82123	3	Lys629 (N-term)		✓			+++	6.46 ± 1.68	2.61 ± 1.27	2.47	4.0E-04
[CRM]-^{279}VNC[MSH]ETSC[MSH]VQQPP C[MSH]FPLK^{295}	1037.42334	2	Cys281					++	11.04 ± 1.57	4.97 ± 1.73	2.22	1.7E-05
			Cys285					+++				
			Cys291					++				
			Val279 (N-term)		✓			++				
[CRM]-^{490}IANVFTNAFR^{499}	598.31458	2	Ile490 (N-term)		✓			+++	45.66 ± 6.94	20.58 ± 6.98	2.22	2.1E-05
[CRM]-^{406}SSEMPELTSMHTLLLR^{421}	944.46881	2	Ser406 (N-term)		✓			+++	10.15 ± 4.72	4.67 ± 2.18	2.17	1.6E-02
[CRM]-^{692}QALAQISLPR^{701}	570.33020	2	Gln692 (N-term)		✓			+++	29.16 ± 4.78	13.47 ± 4.87	2.17	5.5E-05
[CRM]-^{305}NQADC[MSH]IPFFR^{314}	650.28406	2	Cys309					+++	18.89 ± 2.52	8.88 ± 2.84	2.13	1.5E-05
			Asn305 (N-term)		✓			+++				
[CRM]-^{448}KIVGAMVQIITYR^{460}	512.29730	3	Lys448 (N-term)		✓			++	10.84 ± 4.09	5.18 ± 2.03	2.09	6.6E-03
[CRM]-^{604}RFC[MSH]GLPQPETVGQLGTVLR^{622}	720.70660	3	Cys606					++	12.18 ± 1.42	5.82 ± 2.37	2.09	5.5E-05
			Arg604 (N-term)		✓			+++				
[CRM]-^{560}QNQIAVDEIR^{569}	614.81769	2	Gln560 (N-term)		✓			+++	20.39 ± 7.55	9.91 ± 4.50	2.06	8.3E-03
[CRM]-^{572}LFEQVM[Oxi]R^{578}	491.24475	2	Met577	✓				+++	17.20 ± 3.91	8.41 ± 2.24	2.04	2.4E-04
			Leu572 (N-term)		✓			+++				
[CRM]-^{702}IIC[MSH]DNTGITTVSK^{714}	727.35492	2	Cys704					+++	19.06 ± 2.97	9.41 ± 2.96	2.03	5.4E-05
			Ile702 (N-term)		✓			+++				
[CRM]-^{490}IANVFTN[Dea]AFR^{499}	598.80658	2	Asn496			✓		+++	8.28 ± 1.62	4.13 ± 1.30	2.01	1.9E-04
			Ile490 (N-term)		✓			+++				
[CRM]-^{579}IGLDLPALNMQR^{590}	692.37427	2	Ile579 (N-term)		✓			+++	32.43 ± 4.91	16.13 ± 4.97	2.01	4.8E-05
[CRM]-^{593}DHGLPGYNAWR^{603}	664.81018	2	Asp593 (N-term)		✓			+++	18.17 ± 3.06	9.11 ± 3.41	1.99	2.1E-04
^{277}TGVNC[MSH]ETSC[MSH]VQQPP C[MSH]FPLK^{295}	730.30573	3	Cys281					+++	53.87 ± 15.48	27.64 ± 11.89	1.95	4.0E-03
			Cys285					+++				
			Cys291					+++				

(*continued on next page*)

Table 2 (*continued*)

Confirmed sequence of MPO peptides	Theoretical *m/z* of precursor ion	Precursor charges	Amino acid sites	Focused PTMs				Reliability of the modification sites*	Average ± SD of normalized intensity ($\times 10^{-5}$)		MPO-AAV/HC	*p*-values
				Oxidation	Carbamoylation	Deamidation	Glycosylation		MPO-AAV ($n = 7$)	HC ($n = 7$)		
[CRM]-^{448}KIVGAMVQIITYR^{460}	767.94232	2	Lys448 (N-term)		✓			+++	14.66 ± 7.74	7.69 ± 3.34	1.91	4.9E-02
^{277}TGVNC[MSH]ETSC[MSH]VQQPP C[MSH]FPLK^{295}	1094.95496	2	Cys281 Cys285 Cys291					+++ +++ +++	38.55 ± 12.09	20.42 ± 10.00	1.89	9.9E-03
[CRM]-^{537}VVLEGGIDPILR^{548}	662.38519	2	Val537 (N-term)		✓			+++	29.05 ± 5.89	15.53 ± 6.00	1.87	1.1E-03
[CRM]-^{449}IVGAMVQIITYR^{460}	703.89484	2	Ile449 (N-term)		✓			+++	2.11 ± 0.66	1.17 ± 0.71	1.80	2.6E-02
^{278}GVNC[MSH]ETSC[MSH]VQQPP C[MSH]FPLK^{295}	696.62317	3	Cys281 Cys285 Cys291					+++ +++ +++	81.64 ± 15.89	46.20 ± 16.96	1.77	1.7E-03
[CRM]-^{396}IPC[MSH]FLAGDTR^{405}	591.27570	2	Cys398 Ile396 (N-term)		 ✓			+++ +++	35.71 ± 6.72	20.19 ± 7.58	1.77	1.6E-03
[CRM]-^{175}TITGMC[MSH]NNR^{183}	549.72809	2	Cys180 Thr175 (N-term)		 ✓			+++ +++	8.12 ± 2.38	4.62 ± 2.50	1.76	2.0E-02
^{249}SLMFMQW[Kyn]GQLLDHDLDFTPEPAAR^{272}	941.44739	3	Trp255	✓				++	482.71 ± 117.62	289.91 ± 148.28	1.67	1.9E-02
^{278}GVNC[MSH]ETSC[MSH]VQQPP C[MSH]FPLK^{295}	1044.43115	2	Cys281 Cys285 Cys291					+++ +++ +++	32.59 ± 6.65	19.92 ± 6.74	1.64	4.1E-03
[CRM]-^{461}DYLPLVLGPTAMR^{473}	744.89758	2	Asp461 (N-term)		✓			+	22.42 ± 8.43	13.74 ± 4.87	1.63	3.6E-02
^{678}FW[2Ox]WENEGVFSM[Oxi]QQR^{691}	946.40704	2	Trp679 Met688	✓ ✓				+++ +	32.22 ± 7.43	22.11 ± 6.83	1.46	2.1E-02
^{678}FWWENEGVF[Oxi]SMQQR^{691}	930.41211	2	Phe686	✓				+++	1060.56 ± 159.46	737.97 ± 316.82	1.44	3.3E-02
^{605}F[2Ox]C[MSH]GLPQPETVGQLGTVLR^{622}	996.99768	2	Phe605 Cys606	✓				++ ++	32.88 ± 2.68	24.07 ± 7.50	1.37	1.3E-02
^{719}M[2Ox]SNSYPR[CRM]DFVN C[MSH]STLPALNLASWR^{742}	955.10352	3	Met719 Arg725 Cys730	✓	 ✓			+ +++ +++	47.57 ± 4.98	34.87 ± 2.13	1.36	4.6E-05
[+1K]-^{657}VGPLLAC[MSH]IIGTQFR^{670}	554.64581	3	Cys663 Val657 (N-term)					++ ++	10.17 ± 2.36	7.95 ± 1.03	1.28	4.1E-02
[CRM]-^{605}FC[MSH]GLPQPETVGQLGTVLR^{622}	1002.50568	2	Cys606 Phe605 (N-term)		 ✓			+++ +++	23.64 ± 2.61	18.64 ± 4.77	1.27	3.2E-02
[1Ac]-^{719}MSNSYPRDFVNC[MSH] STLPALNLASWR^{742}	944.10852	3	Cys730 Met719 (N-term)					++ +	483.63 ± 45.57	400.06 ± 31.82	1.21	1.8E-03
^{678}FWWENEGVFSM[Oxi]QQR^{691}	930.41211	2	Met688	✓				+++	476.32 ± 43.06	400.69 ± 35.80	1.19	3.8E-03
[AAS]-^{249}SLMFMQWGQ[Dea] LLDHDLDFTPEPAAR^{272}	949.11566	3	Gln257 Ser249 (N-term)			✓		+ ++	16.02 ± 1.96	13.57 ± 2.14	1.18	4.6E-02
^{406}SSEM[2Ox]PELTSMHTLLLR^{421}	626.30963	3	Met409	✓				+++	67.66 ± 9.22	59.52 ± 3.25	1.14	4.8E-02

(*continued on next page*)

Table 2 (*continued*)

Confirmed sequence of MPO peptides	Theoretical *m/z* of precursor ion	Precursor charges	Amino acid sites	Focused PTMs				Reliability of the modification sites*	Average ± SD of normalized intensity ($\times 10^{-5}$)		MPO-AAV/HC	*p*-values
				Oxidation	Carbamoylation	Deamidation	Glycosylation		MPO-AAV ($n = 7$)	HC ($n = 7$)		
^{490}IANVFTN[Dea]AFR^{499}	577.30365	2	Asn496			✓		+++	887.50 ± 83.57	992.50 ± 52.32	−1.12	1.6E-02
^{537}VVLEGGIDPILR^{548}	640.88226	2	none					NA	2145.56 ± 234.71	2395.30 ± 188.51	−1.12	4.9E-02
^{490}IAN[Dea]VFTN[Dea]AFR^{499}	577.79565	2	Asn492			✓		+++	506.56 ± 66.58	588.55 ± 26.96	−1.16	1.1E-02
			Asn496			✓		+++				
^{726}DFVN[HNc]C[MSH]STLPALN[Dea]LASWR^{742}	719.67120	3	Asn729				✓	+++	12.74 ± 3.73	16.79 ± 2.50	−1.32	3.4E-02
			Cys730					+++				
			Asn737			✓		+++				
^{726}DFVNC[MSH]S[HNc]TLPALN[Dea]LASWR^{742}	719.67120	3	Cys730					+++	13.22 ± 3.86	17.49 ± 2.64	−1.32	3.3E-02
			Ser731				✓	+				
			Asn737			✓		+++				
^{463}LPLVLGPTAMR^{473}	584.34955	2	none					NA	7.82 ± 2.57	12.80 ± 5.22	−1.64	4.3E-02
^{303}IKNQADCI[1Me]PFFR^{314}	733.38464	2	Ile310					+	3.57 ± 2.30	6.05 ± 1.71	−1.69	4.1E-02
^{355}N[HNc]MSNQLGLLAVNQR^{368}	880.95160	2	Asn355				✓	+++	7.03 ± 3.70	15.09 ± 5.87	−2.15	9.6E-03
^{355}NMS[HNc]NQLGLLAVNQR^{368}	880.95160	2	Ser357				✓	+++	6.53 ± 3.50	15.17 ± 4.24	−2.32	1.3E-03
^{512}LDNRYQPMEPNPR^{524}	543.93158	3	none					NA	3.83 ± 1.86	8.89 ± 3.70	−2.32	7.2E-03

PTM abbreviations followed Unimod: AAS, acetaldehyde +26; CRM, carbamoylation; Dea, deamidation; HNc, N-Acethylhexosamine; Kyn, tryptophan oxidation to kynurenin; MSH, beta-methylthiolation; Oxi, oxidation; +1K, addition of lysine due to transpeptidation; 1Ac, acetylation; 1Me, methylation; 2Ox, dioxidation.

* Reliability of the modification sites: +++, the modified amino acids were identified from the MS/MS spectra; ++, the modified amino acids were not identified from the MS/MS spectra, however, they were identified from already-known information of amino acids that undergo the PTM types; +, the modified amino acids were not identified both from the MS/MS spectra and already-known information. NA, not applicable.

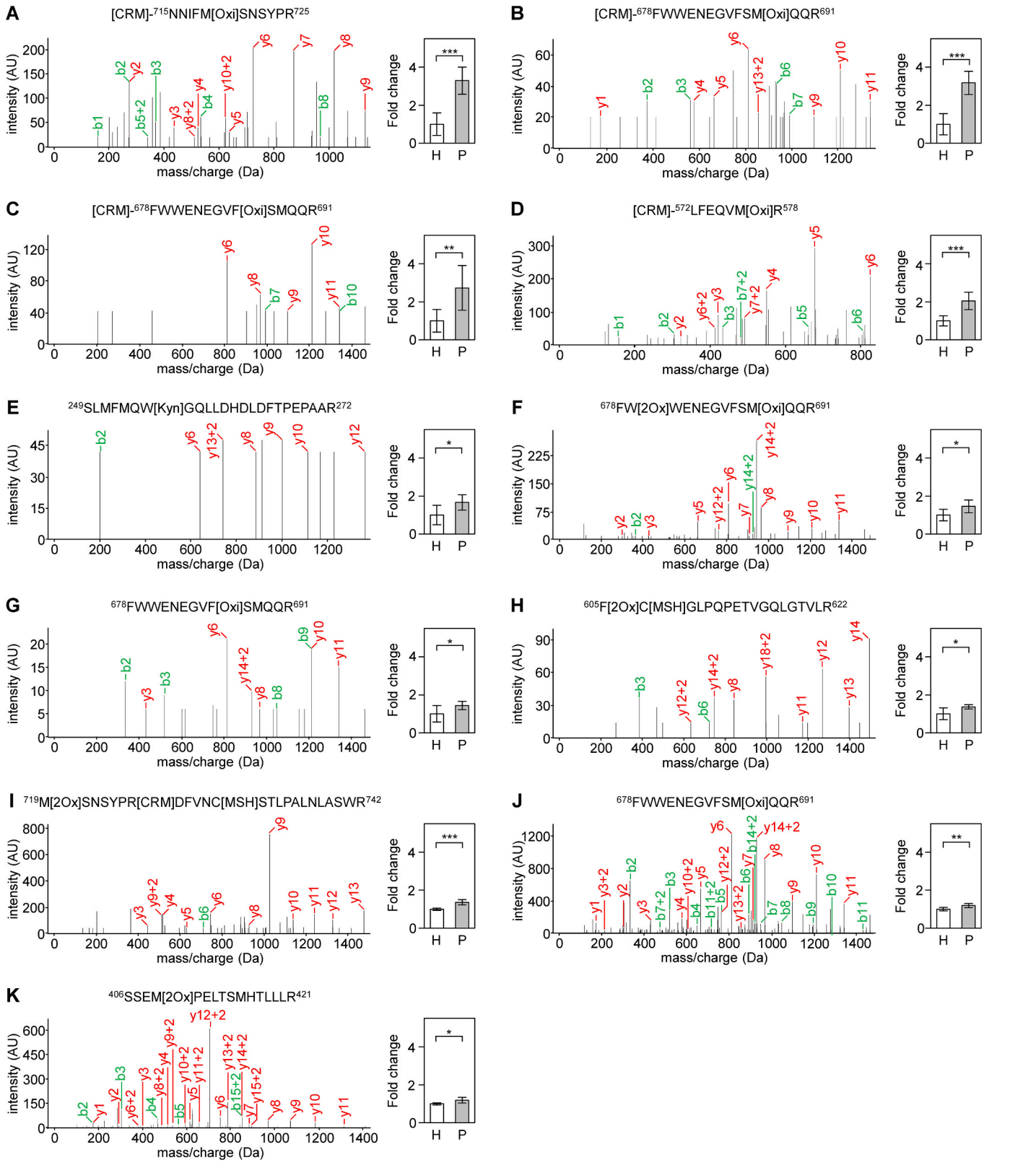


Fig. 2. Oxidatively modified MPO peptides increased in MPO-AAV patients.
Eleven oxidatively modified peptides were found to be increased in the MPO-AAV group relative to the HC group in the SWATH mode. The results of MS/MS analysis and the quantitative values are shown. The normalized quantitative values of the HC group are shown as 1.0. CRM, carbamoylation; Kyn, tryptophan oxidation to kynurenin; MSH, β-methylthiolation; Oxi, oxidation; 2Ox, dioxidation. H, HC group ($n = 7$); P, MPO-AAV group ($n = 7$). *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

Table 3
mMPO-derived peptides whose quantitative values were increased more than 30-folds in the H_2O_2-treated mMPO relative to the non-treated mMPO.

Confirmed sequence of MPO peptides	Theoretical *m/z* of precursor ion	Precursor charges	Amino acid sites	Including oxidative modification	Reliability of the modification sites	Average ± SD of the quantitative values*** ($\times 10^{-7}$)		H_2O_2-treated /non-treated	*p*-values
						H_2O_2-treated	non-treated		
^{103}KLQPLW[2Ox]PR^{110}	535.31128	2	Trp108	✓	+++	1409.13 ± 220.81	9.09 ± 5.06	154.98	3.9E-04
^{435}DYLPLVLGPAAM[2Ox]K^{447}	710.38123	2	Met446	✓	+++	3467.56 ± 69.85	22.76 ± 13.38	152.35	1.2E-07
^{603}KLMAQY[Oxi]GTPNNIDIW [2Ox]M[Oxi] GGVSEPLEPN[Dea]GR^{630}	1051.49365	3	Tyr608 Trp617 Met618 Asn628	✓	+++ +++ +++ +++	1931.48 ± 88.16	18.46 ± 6.91	104.62	3.0E-06
^{648}DGDRFWW[1Br] ENPGVFSK^{662}	639.92334	3	Trp654		+++	1888.52 ± 94.66	19.71 ± 11.45	95.82	4.5E-06
^{89}AADY[1Br]LHVALDLLK^{101}	507.24335	3	Tyr92		+++	2999.70 ± 304.11	33.49 ± 11.85	89.57	7.2E-05
^{435}DYLP[PGP]LVLGPAAM [Oxi]K^{447}	709.37341	2	Pro438 Met446	✓	++ +	2176.33 ± 58.79	24.71 ± 2.64	88.09	3.7E-07
^{435}D[Dhy]Y[2Ox] LPLVLGPAAM[Oxi]K^{447}	709.37341	2	Asp435 Tyr436 Met446	✓	++ ++ +	1229.24 ± 44.59	13.96 ± 1.54	88.05	1.2E-06
^{504}VFFASW[1Br]R^{510}	495.69522	2	Trp509		+++	1338.40 ± 57.97	15.91 ± 7.40	84.14	2.5E-06
^{652}FWW[1Br]ENPGVFSK^{662}	737.79309	2	Trp654		+++	9065.07 ± 34.82	113.27 ± 61.71	80.03	2.6E-09
^{553}IGLDLPALNM[2Ox]QR^{564}	686.86627	2	Met562	✓	+++	3532.35 ± 75.34	47.16 ± 22.36	74.91	1.7E-07
[AAS]-^{553}IGLDLPALNM[Oxi] QR^{564}	691.87665	2	Met562 Ile553 (N-term)	✓	+++ +++	8901.94 ± 830.30	121.52 ± 100.82	73.25	5.4E-05
^{603}KLM[Oxi]AQYGTPN[Dea] NIDIW[1Br] MGGVSEPLEPN[Dea]GR^{630}	796.59949	4	Met605 Asn612 Trp617 Asn628	✓	+ + ++ +++	10,229.51 ± 331.78	164.38 ± 151.85	62.23	1.1E-06
^{603}KLM[Oxi]AQYGTPN[Dea] NIDIWM[Oxi] GGVSEPLEPN[Dea]GR^{630}	781.12061	4	Met605 Asn612 Met618 Asn628	✓	+ +++ +++ +++	10,229.51 ± 331.78	164.46 ± 150.16	62.20	1.1E-06
^{383}M[Oxi]PELTSM[Oxi] HTLFVR^{395}	797.39185	2	*Met383 Met389	✓	+++ +++	247.77 ± 16.48	4.00 ± 4.09	61.91	1.6E-05
^{603}KLM[Oxi]AQYGTPNNIDIW [1Br]MGGVSEPLEP N[Dea]GR^{630}	796.35345	4	Met605 Trp617 Asn628	✓	+ ++ +++	8008.64 ± 124.64	129.63 ± 102.10	61.78	1.2E-07
^{435}DYLPLVLGPAAM[Oxi]KK^{448}	511.28995	3	Met446	✓	+++	84,566.49 ± 1504.33	1574.61 ± 105.66	53.71	7.3E-08
^{438}PLVLGPAAM[Oxi]KK^{448}	570.84412	2	Met446	✓	+++	3319.78 ± 57.93	64.26 ± 15.31	51.66	7.6E-08
^{223}ALM[Oxi]FM[Oxi]QW[Oxi] GQFLD[MSH] HDITLTPEPATR^{246}	971.45203	3	Met225 Met227 *Trp229 Asp234	✓	+++ +++ +++ +++	2182.21 ± 169.52	43.40 ± 17.90	50.28	2.7E-05
^{603}KLM[Oxi] AQYGTPNNIDIWM[Oxi] GGVSEPLEP N[Dea]GR^{630}	1040.83032	3	Met605 Met618 Asn628	✓	+++ +++ +++	11,202.21 ± 326.95	227.51 ± 172.87	49.24	8.6E-07
^{618}M[Oxi]GGVSEPLEPN[Dea] GR^{630}	680.31409	2	Met618 Asn628	✓	+++ +++	674.84 ± 68.64	14.13 ± 1.42	47.76	7.6E-05
^{380}SSEM[Oxi]PELTSM[Oxi] HTLF^{393}	821.36035	2	*Met383 Met389	✓	+++ +++	663.67 ± 16.75	14.03 ± 3.73	47.30	3.2E-07
^{172}WLPAEYEDGVSM[Oxi] PFGW[Ntr]TP[Oxi] GVN[Dea]R^{193}	862.71289	3	Met183 Trp187 Pro189 Asn192	✓	+++ +++ +++ +++	9623.92 ± 300.86	213.45 ± 117.04	45.09	9.2E-07
^{380}SSEM[Oxi]PELTSM[Oxi] HTLFVR^{395}	632.96588	3	*Met383 Met389	✓	+++ +++	60,615.37 ± 2612.34	1417.58 ± 173.66	42.76	2.5E-06
[1Ac]-^{168}AFVRW[HKy] LPAEYEDGVSM[Oxi] PFGWTPGVNR^{193}	1020.48340	3	Trp172 Met183 Ala168 (N-term)	✓	++ +++ ++	23,207.42 ± 594.70	555.10 ± 578.88	41.81	1.2E-06
^{603}KLM[Oxi]AQYGTPNNIDIW [1Br]MGGVSEPLEP N[Dea]GR^{630}	1061.46887	3	Met605 Trp617 Asn628	✓	+++ ++ +++	9652.62 ± 69.54	231.39 ± 175.68	41.72	1.1E-07

(*continued on next page*)

Table 3 (*continued*)

Confirmed sequence of MPO peptides	Theoretical *m/z* of precursor ion	Precursor charges	Amino acid sites	Including oxidative modification	Reliability of the modification sites	Average ± SD of the quantitative values*** ($\times 10^{-7}$)		H_2O_2-treated /non-treated	*p*-values
						H_2O_2-treated	non-treated		
^{603}KLM[Oxi]AQYGTPNNIDIW[Kyn]M[Oxi]GGVSEPLEPN[Dea]GR^{630}	781.87329	4	Met605 Trp617 Met618 Asn628	✓	+++ ++ +++ +++	8846.83 ± 157.28	236.70 ± 30.00	37.38	8.0E-08
^{603}KLMAQY[Oxi]GTPNNIDIWM[Oxi]GGVSEPLEPN[Dea]GR^{630}	1040.83032	3	Tyr608 Met618 Asn628	✓	+++ +++ +++	11,179.36 ± 159.69	299.58 ± 60.82	37.32	4.1E-08
^{689}NNIFM[Oxi]SNT[Dhy]YPR^{699}	452.21179	3	*Met693 Thr696	✓	+++ +++	328.96 ± 24.11	8.94 ± 4.12	36.81	2.2E-05
^{553}IGLDLPALNM[Oxi]QR^{564}	678.86884	2	Met562	✓	+++	211,613.50 ± 2646.10	5963.53 ± 938.41	35.48	2.3E-08
^{370}IPC[MSH]FLAGDM[2Ox]R^{379}	600.76416	2	Cys372 Met378	✓	+++ +++	2918.35 ± 115.30	86.52 ± 4.35	33.73	1.8E-06
[AAS]-^{435}DYLPLVLGPAAM[Oxi]KK^{448}	519.96179	3	Met446 Asp435 (N-term)	✓	+++ ++	1885.17 ± 134.22	59.07 ± 15.85	31.91	2.0E-05
^{435}DYLPLVLGPAAM[Oxi]K^{447}	702.38379	2	Met446	✓	+++	139,412.13 ± 2046.43	4459.32 ± 897.18	31.26	5.0E-08
[MSH]-^{648}DGDRFW[Oxi]WENPGVFSK^{662}	634.61401	3	*Trp653 Asp648 (N-term)	✓	+ ++	2233.86 ± 21.80	72.73 ± 53.43	30.71	3.4E-07
[AAS]-^{546}LFEQVM[Oxi]R^{552}	482.74966	2	**Met551 Leu546 (N-term)	✓	+++ +++	4345.17 ± 68.75	267.73 ± 210.05	16.23	5.7E-06

The reliability of PTM sites: +++, the modified amino acids were identified from the MS/MS spectra; ++, the modified amino acids were not identified from the MS/MS spectra but identified from the already-known information of amino acids that undergo PTM types; +, the modified amino acids were not identified from the MS/MS spectra nor already-known information.

PTM abbreviations followed Unimod: AAS, acetaldehyde +26; Dea, deamidation; Dhy, dehydration; HKy, tryptophan oxidation to hydroxykynurenin; Kyn, tryptophan oxidation to kynurenin; MSH, beta-methylthiolation; Ntr, oxidation to nitro; Oxi, oxidation; PGP, proline oxidation to pyroglutamic acid; 1Ac, acetylation; 1Br, bromination; 2Ox, dioxidation.

* Amino acids corresponding to the human MPO amino acids whose oxidative modification was increased in the MPO-AAV group relative to the healthy group.

** Met551 corresponded to the Met577 in human MPO whose oxidative modification was increased in the MPO-AAV group relative to the healthy group, although the increase of this mMPO-derived peptide was less than 30-fold in H_2O_2-treated mMPO relative to non-treated mMPO.

*** Average and standard deviation (SD) of the quantitative values measured in triplicate are shown.

4. Discussion

In the present study, we hypothesized that an aberrant PTM that induces MPO-ANCA production is added to neutrophil MPO proteins in patients with MPO-AAV. PTMs of neutrophil MPO proteins obtained from the MPO-AAV and HC groups were comprehensively quantified using a SWATH-MS analysis. We found that MPO peptides containing oxidative modifications of Met, Phe, and Trp significantly increased in the MPO-AAV group relative to the HC group (Table 2). Some of these amino acids are located on the surface of MPO molecule (Supplementary Fig. 7). This suggests that the increased oxidative modifications of these amino acids may change the antigenicity of MPO.

To test the effects of increased oxidative modifications on the immunogenicity of MPO proteins, mice were immunized with mMPO proteins oxidized by H_2O_2-treatment and the production of anti-H_2O_2-treated and anti-non-treated mMPO antibodies was examined. Oxidative modifications were markedly increased in H_2O_2-treated mMPO relative to non-treated mMPO (Table 3). Notably, 4 amino acid sites with increased oxidative modifications in mMPO corresponded to the amino acid sites whose kynurenylation, dioxidation, or oxidation were increased in the MPO-AAV group (Supplementary Fig. 5 and Table 3). Immunization of mice with H_2O_2-treated mMPO resulted in the production of anti-H_2O_2-treated mMPO antibodies (Fig. 3BD, left). ROS-mediated oxidative modifications of amino acid side chains cause novel antigenicity at these sites [34]. Furthermore, ROS are known to induce responses in proteins, causing their unfolding and exposing their cryptic epitopes [17]. In this study, treatment with H_2O_2 was considered to similarly induce oxidative modifications of amino acid side chains or conformational changes in MPO, resulting in the formation of a novel epitope recognized by the immune system.

Notably, antibodies against non-treated mMPO were also produced in mice immunized with H_2O_2-treated mMPO (Fig. 3 CE, left). This production of antibodies against the self-protein may be due to either of the following reasons: 1) the newly generated highly oxidized sites of MPO were recognized as epitopes by T and B cells, leading to the production of anti-oxidized MPO antibodies (Supplementary Fig. 8), which cross-reacted to the same sites as the normal MPO due to molecular mimicry; 2) the above T cells activated autoreactive B cells that had been in an anergic state according to peripheral tolerance (Supplementary Fig. 8) [35,36]. Given the high antigen specificity of antibodies, the latter was considered likely.

The SWATH results also revealed that *N*-linked glycosylation at Asn355 and Asn729, which are located on the surface of MPO molecules, was reduced in the patients with MPO-AAV (Table 2). Recently, it has been reported that the glycans at Asn355 and Asn391 in MPO were partially deglycosylated by Endoglycosidase H treatment and that deglycosylation increased hypochlorous acid production by MPO [25]. The enzymatic activity of MPO may be increased in MPO-AAV owing to the decrease in the glycosylation.

A decrease in not only *N*-linked glycosylation (Asn355 and Asn729) but also in *O*-linked glycosylation (Ser357 and Ser731) was observed in MPO proteins from the MPO-AAV patients (Table 2). At these sites, epitopes normally covered by glycans may be exposed to the immune system owing to the decrease in glycosylation. The exposed epitopes may be recognized by T cells and B cells reactive to non-self-antigens, leading to the production of MPO-ANCA.

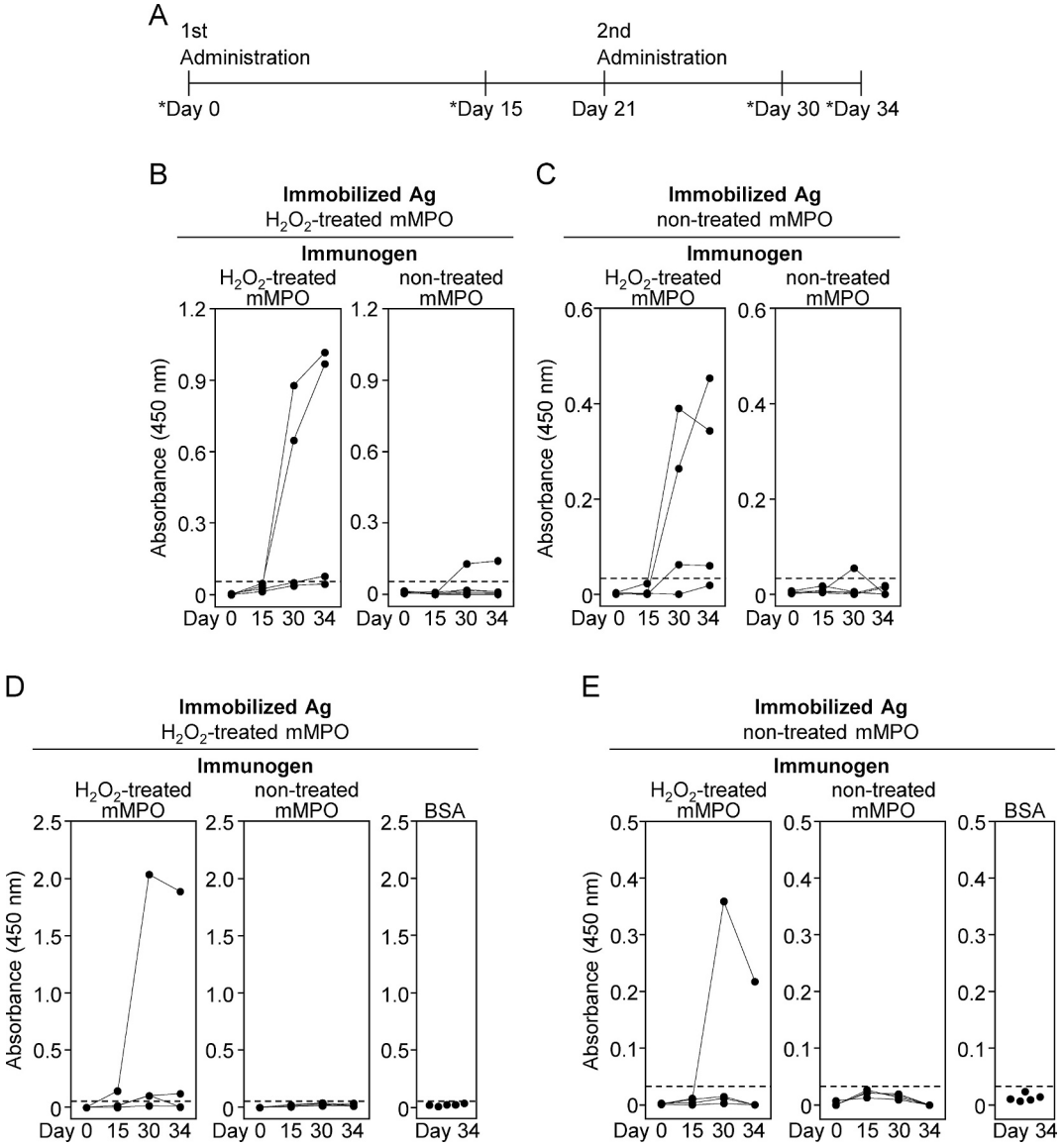


Fig. 3. Induction of anti-MPO antibodies by immunization with H_2O_2-treated mMPO.
(A) Schedule of mouse immunizations and plasma sampling. B6 mice were immunized with either H_2O_2-treated mMPO or non-treated mMPO on days 0 and 21. *Plasma samples were obtained on days 0, 15, 30, and 34.
(B) An ELISA result in which H_2O_2-treated mMPO was used as the solid-phase antigen, and plasma samples of the 1st group mice immunized with H_2O_2-treated mMPO ($n = 4$) or non-treated mMPO ($n = 5$) were used as the primary antibodies.
(C) An ELISA result in which non-treated mMPO was used as the solid-phase antigen. The plasma samples of the 1st group mice were used as the primary antibodies.
(D) An ELISA result in which H_2O_2-treated mMPO was used as the solid-phase antigen, and plasma samples of the 2nd group mice immunized with H_2O_2-treated mMPO ($n = 4$), non-treated mMPO ($n = 4$), or BSA ($n = 5$) were used as the primary antibodies.
(E) An ELISA result in which non-treated mMPO was used as the solid-phase antigen. The plasma samples of the 2nd group mice were used as primary antibodies.
The dotted lines in B-E show the mean + 3SD of the day 34 plasma samples from BSA-immunized mice. Values that exceeded the cut-off values were determined to be positive.

Table 4
Increase of the anti-H_2O_2-treated and anti-non-treated mMPO antibody production ratios by immunization of H_2O_2-treated mMPO.

	Immobilized Ag: H_2O_2-treated mMPO				Immobilized Ag: non-treated mMPO			
	Immunogen				Immunogen			
Reaction with immobilized Ag	H_2O_2-treated mMPO	non-treated mMPO	Total	*p*-value	H_2O_2-treated mMPO	non-treated mMPO	Total	*p*-value
positive	5	1	6		4	0	4	
negative	3	8	11	< 0.05	4	9	13	< 0.05
Total	8	9	17		8	9	17	

Plasma samples were collected on day 34, and antibody titers to H_2O_2-treated mMPO and non-treated mMPO were measured by an ELISA. The mean + 3SD values of BSA-immunized mice were used as the cutoff values to determine whether the antibodies were positive or negative. Differences in the respective antibody production ratios between H_2O_2-treated mMPO- and non-treated mMPO-immunized mice were determined using Fisher's exact test.

N-terminal carbamoylation was found in most of the MPO peptides that increased in the MPO-AAV group relative to the healthy group (Table 2). The N-terminal carbamoylation of peptides is an artificial modification that occurs in the presence of urea [37]. Urea was used in our experiments in the steps of reductive alkylation and tryptic digestion. Because almost all the carbamoylated N-termini of MPO peptides were trypsin cleavage sites, the N-terminal carbamoylation in our results was considered an artificial modification caused by the experimental procedures.

The limitations associated with the present study were as follows: First, the number of the MPO-AAV patients and HC subjects analyzed using SWATH-MS was small. The results should be validated in a cohort that including a large number of patients with MPO-AAV and healthy subjects. Second, in the mouse experiment, the T cell epitopes that drive the production of anti-normal MPO antibodies remain unclear. Further studies are required to clarify this point.

5. Conclusions

In this study, SWATH showed that the oxidative modifications of Met, Phe, and Trp in MPO proteins derived from neutrophils were increased in MPO-AAV patients compared to those in healthy subjects. Immunization of mice with oxidized mMPO induced not only anti-oxidized MPO antibodies but also anti-unoxidized MPO autoantibodies, suggesting that the increased oxidatively modifications of MPO may trigger MPO-ANCA production. The findings of this study are considered to help us better understand the etiology of MPO-AAV.

Funding

This study was financially supported by Japan society for the promotion of science (JPSP) KAKENHI Grant Number JP 23K07915.

CRediT authorship contribution statement

Masaaki Sato: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Kouhei Nagai:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Toshiyuki Sato:** Writing – review & editing, Investigation. **Ryo Yoshimoto:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Yuto Shibano:** Writing – review & editing, Validation, Investigation. **Minori Shibahara:** Writing – review & editing, Validation, Investigation. **Haruka Satokawa:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Masayuki Anzai:** Writing – review & editing, Validation, Methodology, Investigation. **Teisuke Uchida:** Writing – review & editing, Resources, Investigation. **Atsuhiro Tsutiya:** Writing – review & editing, Investigation. **Yukiko Takakuwa:** Writing – review & editing, Resources, Investigation. **Kazuki Omoteyama:** Writing – review & editing, Investigation. **Mitsumi Arito:** Writing – review & editing, Investigation. **Naoya Suematsu:** Writing – review & editing, Investigation. **Seido Ooka:** Writing – review & editing, Resources, Investigation. **Kimito Kawahata:** Writing – review & editing, Resources, Investigation. **Tomohiro Kato:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Conceptualization. **Manae S. Kurokawa:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no competing interests.

Acknowledgement

The authors indebted to Ms. Michiyo Katano for her technical assistance.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jprot.2025.105412.

Data availability

The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE [38] partner repository with the dataset identifier PXD050920.

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